

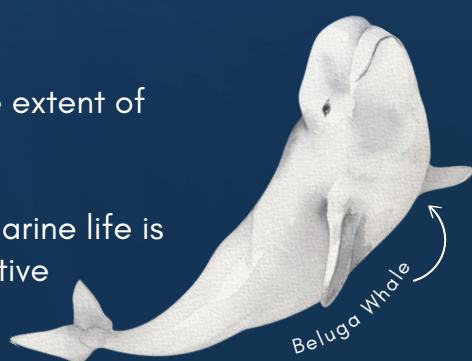
ALL NOISY ON THE ARCTIC WATERFRONT

THE PROBLEM OF NOISE POLLUTION IN CANADA'S ARCTIC WATERS

What is noise pollution?

Simply put, noise pollution is the excess of unwanted sound, to the extent of having detrimental effects on surrounding people and wildlife. [6]

Noise pollution can occur on land, as well as underwater, where marine life is often reliant on sound to navigate its surroundings. Thus, the negative effects of noise pollution are especially pronounced in the quiet waters of Canada's Arctic. [4]



How does anthropogenic (human) noise reach the Arctic?

An increased demand for commercial shipping has led to more cargo ships travelling through these waters. [4]

Sea ice melting due to global warming is allowing more ships into the Arctic—cargo ships travelling through these waters increased by **25%** between **2013-2019** alone. [4]

Over the last **25 years**, the distance travelled by commercial ship routes in Canada's Arctic has increased by almost **3x** [8]; noise levels have gone up almost **10x**. [9]



DID YOU KNOW?

Large cargo ships can be louder (up to **190 decibels**) [2] than a jet at takeoff (about **150 decibels**) [5]

Melting sea ice is not just opening up the Arctic to noisy human activity—the once tightly-packed areas of ice are also losing their capacity to protect the water from loud sounds caused by wind and choppy waters. [4]

Main noise polluters

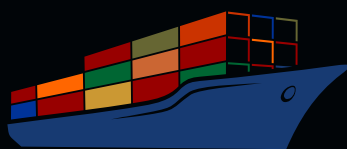
- Ships (cargo, cruise, icebreakers, military) [4]

- Development (oil drilling, construction) [2]

- Research activities (sonars, seismic air guns which blast air and measure the echo to map the ocean floor) [3]



How loud are they?
Scan to hear the noises of cargo ships, air guns, and the Arctic marine life that endure them! [8]



How is noise pollution harmful?

1

COMMUNICATION INTERFERENCE

Arctic mammals that use sound to communicate, like beluga whales and narwhals, can't hear each other properly because of loud ships [1]

2

MIGRATION & BEHAVIOUR

Migratory routes and home ranges of marine animals like Arctic cod are shifting to avoid ship traffic [4]

3

INJURY & DEATH

Air guns (which can be heard over 500 km away) [3] and shipping noises are so loud they can cause hearing loss, physical injury, and death in marine animals [1]

4

INDIGENOUS COMMUNITIES AT RISK

Indigenous communities in northern Canada are losing access to important Arctic marine animals being driven away by noise pollution [1]

WHAT CAN BE DONE?

The silver lining: Noise pollution in the Arctic can be significantly reduced quickly, since noise doesn't remain in the water like CO2 remains in the air. [4]

Shhh...

DID YOU KNOW?

Cortisol, the "stress hormone", has increased by over **100%** in narwhals exposed to shipping activity near Baffin Island, Canada since 2006. [7]

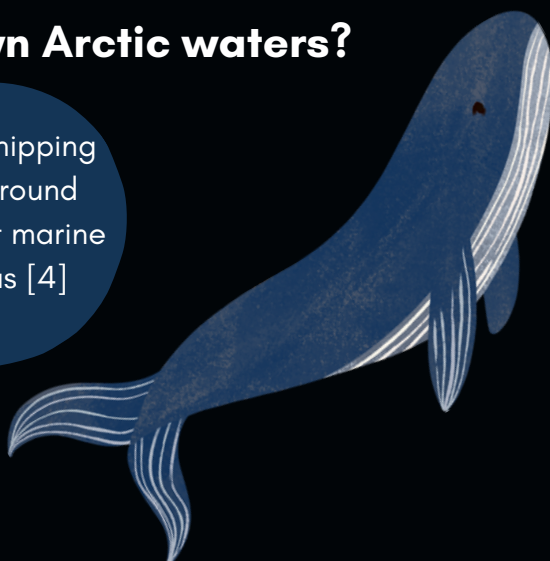


So, how can we quiet down Arctic waters?

Modify ship design to produce less underwater noise [4]

Restrict shipping routes around significant marine life areas [4]

Slow down ship speed [4]



References!



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